Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In fructose, the possible optical isomers are

(A) 12

(B) 8

(C) 16

(D) 4

Q.2

Accumulation of which of the following molecules in the muscles occurs as a result of vigorous exercise ?

(A) Glycogen

(B) Glucose

(C) Pyruvic acid

(D) L-lactic acid

Q.3

Which of the following will not show mutarotation?

(A) Maltose

(B) Lactose

(C) Glucose

(D) Sucrose

Q.4

An organic compound with the formula $C_6H_{12}O_6$ forms a yellow crystalline solid with phenylhydrazine and gives a mixture of sorbitol and mannitol when reduced with sodium. Which among the following could be the compound?

(A) fructose

(B) glucose

(C) mannose

(D) sucrose

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Q.5

When α -D-glucose and β -D-glucose are dissolved in water in two separate beakers I and II respectively and allowed to stand, then -

(A) specific rotation in beaker I will decrease while in II will increase upto a constant value

(B) the specific rotation of equilibrium mixture in two beakers will be different

(C) the equilibrium mixture in both beakers will be leavorotatory

(D) the equilibrium mixture in both beakers will contain only cyclic form of glucose

Q.6

Which of the following pairs can be distinguished by Fehling's solution ?

(A) Glucose and fructose

(B) Glucose and sucrose

(C) Methanal and ethanal

(D) Hydroxypropanone and benzaldehyde.

Q.7

Natural glucose is termed D-glucose because :

(A) $-\mathrm{OH}$ on the second carbon is on the right side in Fischer projection

(B) $-\mathrm{OH}$ on the sixth carbon is on the right side in Fischer projection.

(C) $-\mathrm{OH}$ on the fifth carbon is on the right side in Fischer projection.

(D) It is dextrorotatory.

Q.8

Amylopectin is a polymer of

(A) $\alpha - D-$ glucose

(B) amino acid

(C) $\beta - D-$ glucose

(D) amylase.

Q.9

Glycosidic linkage is actually an :

(A) Carbonyl bond

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(B) Ether bond

(C) Ester bond

(D) Amide bond

Q.10

Which of the following carbohydrates is not related to $(+)$ -glucose?

(A) Amylopectin

(B) Amylose

(C) Inulin

(D) Glycogen

Q.11

Glucose-D has a great tendency to be converted into cyclic isomer. Which two carbon atoms get joined through ' O ' to form this hemiacetal ?

(A) C_1 and C_4

(B) C_1 and C_5

(C) C_1 and C_6

(D) C_2 and C_6

Q.12

The enantiomer of $\alpha-\mathrm{D}-(+)$ -glucose is

(A) $\beta-\mathrm{D}-(+)$ -glucose

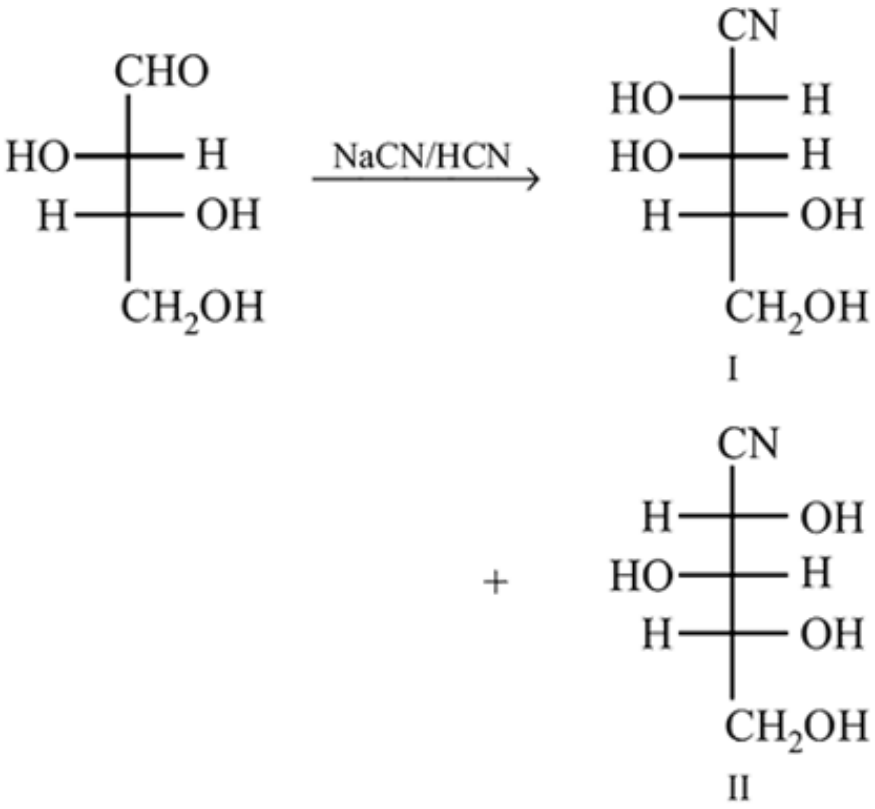
(B) α -D- $(-)$ -glucose

(C) α -L- $(-)$ -glucose

(D) $\beta-\mathrm{L}-(-)$ -glucose

Q.13

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Compounds I and II may be grouped as

(A) diastereomers

(B) epimers

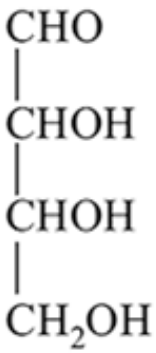
(C) $C-2$ epimers

(D) all of the three.

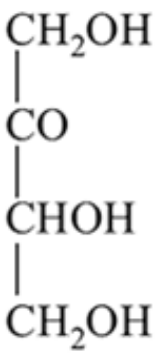
Q.14

Which of the following evolves carbon dioxide, on oxidation with periodate?

(A)



(B)



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(C) Both

(D) None

Q.15

Among the following organic acids, the acid present in rancid butter is:

(A) Pyruvic acid

(B) Lactic acid

(C) Butyric acid

(D) Acetic acid

Q.16

Number of chiral carbons in $\beta - \mathrm{D} - (+) -$ glucose is

(A) There

(B) Four

(C) Five

(D) Six

Q.17

A certain compound gives negative test with ninhydrin and positive test with Benedict's solution. The compound is

(A) A protein

(B) A monosaccharide

(C) A lipid

(D) An amino acid

Q.18

Glucose forms many derivatives. The derivative which will help to prove the furanose structure is

(A) Acetyl

(B) Benzoyl

(C) Osazone

(D) Isopropylidene

Q.19

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The term anomers of glucose refers to

(A) Isomers of glucose that differ in configurations at carbons one and four $(\mathrm{C}-1$ and $\mathrm{C}-4)$

(B) A mixture of (D)-glucose and (L)-glucose

(C) Enantiomers of glucose

(D) Isomers of glucose that differ in configuration at carbon one (C-1)

Q.20

Glucose on oxidation with nitric acid yields

(A) Saccharic acid

(B) Gluconic acid

(C) Oxalic acid

(D) Sorbitol

Answers & Solutions

Q.1 Answer:

8

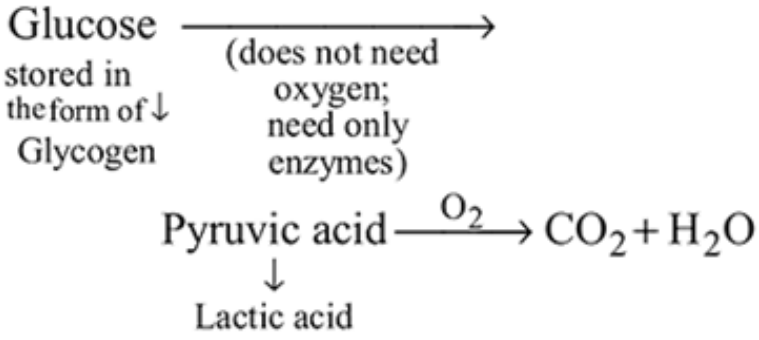
Solution:

Fructose has 3 chiral centres and hence number of optical isomers are $2^3 = 8$

Q.2 Answer:

L-lactic acid

Solution:



During vigorous exercise sufficient oxygen is not available to meet the energy demand so, energy is derived through conversion of pyruvic acid to lactic acid.

Q.3 Answer:

Sucrose

Solution:

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Sucrose does not contain a free aldehydic or ketonic group hence it does not show mutarotation.

Q.4 Answer:

fructose

Solution:

Since the compound forms a yellow crystalline solid, i.e. osazone with phenylhydrazine, it may be an aldohexose or a ketohexose. Further, since on reduction, compound forms a mixture of sorbitol and mannitol, it must be a ketohexose, i.e. fructose. Recall that glucose on reduction gives only one alcohol glucitol (Sorbitol)

Q.5 Answer:

specific rotation in beaker I will decrease while in II will increase upto a constant value

Solution:

α -D-glucose or β -D-glucose when dissolved in water and allowed to stand, following equilibrium is established.

α -D-glucose $\rightleftharpoons$ Open chain form $\rightleftharpoons$ β -D-glucose

$(+111^{\circ})$ $(+19^{\circ})$

Specific rotation of α -form falls, while that of β -form increases until a constant value of $+52.5^{\circ}$ is reached at equilibrium. This phenomenon is known as mutarotation.

Q.6 Answer:

Glucose and sucrose

Solution:

Glucose is a hemiacetal, so in presence of a base (alkaline medium is provided by Fehling's solution) it can develop - CHO group in the form of open chain structure which responds Fehling's solution. Sucrose is a glycoside (acetal), i.e. its hemiacetal OH groups (one due to glucose and another due to fructose) are not free, so it can't attain -CHO group. Hence it will not respond Fehling's solution.

Q.7 Answer:

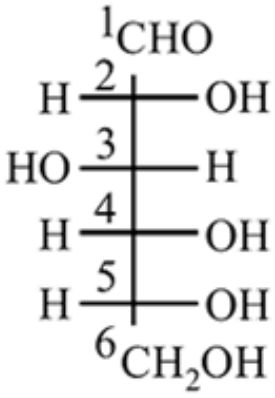
$-\mathrm{OH}$ on the fifth carbon is on the right side in Fischer projection.

Solution:

Fischer gave the prefix "D" to compounds whose bottom chiral has its OH to the right. So natural glucose is called D-glucose or dextrose.

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Structure of D-Glucose :



D-Glucose

Q.8 Answer:

$\alpha - D-$ glucose

Solution:

Amylopectin is a polymer of $\alpha - D$ -glucose.

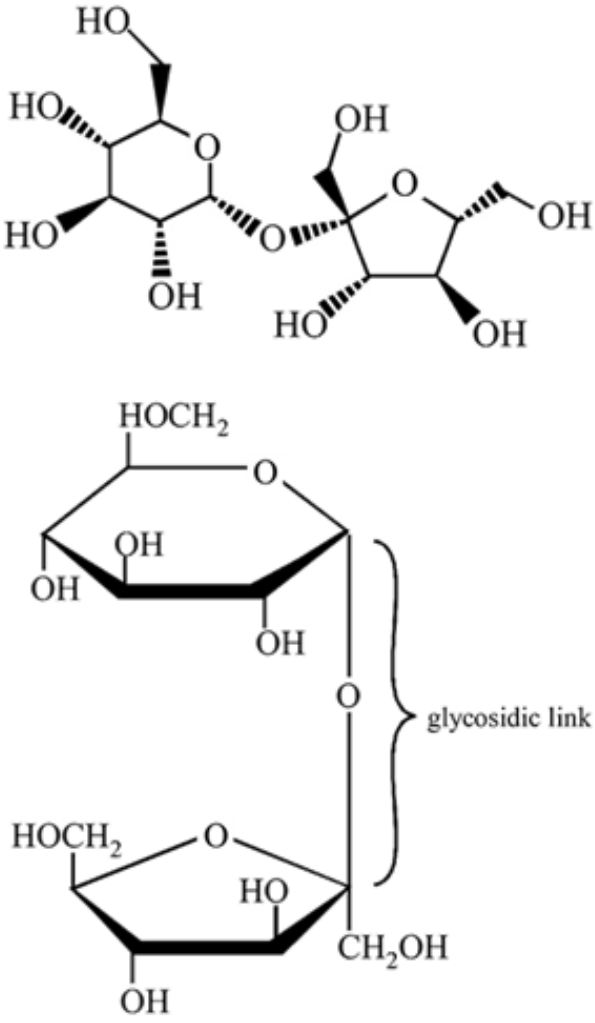
Q.9 Answer:

Ether bond

Solution:

Glycosidic linkage is actually an ether bond as the linkage forming the rings in an oligosaccharide or polysaccharide is not just one bond, but the two bonds sharing an oxygen atom e.g. sucrose

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Q.10 Answer:

Inulin

Solution:

Inulin is a polysaccharide of fructose, while all others are polysaccharides of glucose ; although all the four has same molecular formula, $(C_6H_{10}O_5)_n$.

Q.11 Answer:

C_1 and C_5

Solution:

Glucose exists as glucopyranose (a six membered ring structure) which is formed through C_1 and C_5.

Q.12 Answer:

α -L- $(-)$ -glucose

Solution:

In the D family the more dextrorotatory anomer is named $\alpha-D-$. In the L family the more laevorotatory anomer is named αL

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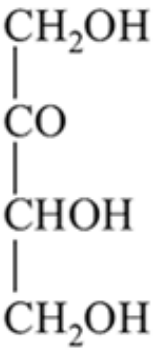
Q.13 Answer:

all of the three.

Solution:

When structures I and II are $C-2$ epimers, it implies that these are epimers and diastereomers too.

Q.14 Answer:



Solution:

A ketonic group having $-OH$ group on both sides is removed as CO_2 during oxidation with periodate.

Q.15 Answer:

Butyric acid

Solution:

Butyric acid also known as butanoic acid is found in milk and butter and is a product of anaerobic fermentation. It has an unpleasant smell and acid taste.

Q.16 Answer:

Five

Solution:

As seen in image above, there are five chiral carbons in β - $D(+)$-glucose.A chiral carbon has four different groups attached. -D-Glucose has no plane of symmetry, so all the carbon atoms are different. There are five chiral carbons.

Q.17 Answer:

A monosaccharide

Solution:

Protein gives blue-violet color with ninhydrin (2,2-dihydroxyindane-1,3-dione)
Carbohydrates gives brown red ppt.with benedict's solution (Alk. $CuSO_4$ +Citrate ions)
A compound which gives a negative test with ninhydrin, it cannot be a protein or an amino acid. As, it gives a positive test with Benedict's solution. So, it must be a monosaccharide but not a lipid.

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Q.18 Answer:

Osazone

Solution:

Glucose and fructose differ in the arrangement of atoms around the $C-1$ and $C-2$ carbon. When both of them react with three molecules of phenylhydrazine, the formation of osazone crystals makes the difference between the carbon atoms in their chain and hence, in such way it helps to prove the fructose structure.

Q.19 Answer:

Isomers of glucose that differ in configuration at carbon one (C-1)

Solution:

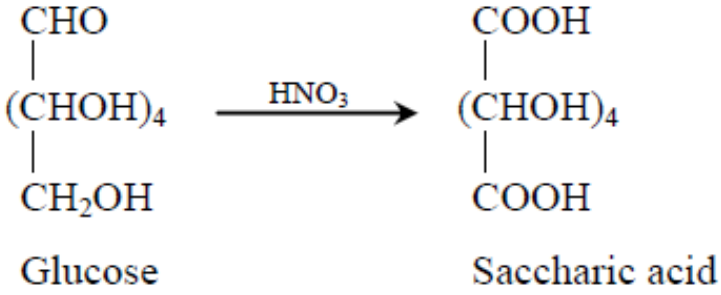
Anomers are such diastereomeric molecules which differ in the configuration of $C-1$ carbon only.

Q.20 Answer:

Saccharic acid

Solution:

Glucose is oxidized by nitric acid and produce saccharic acid. Saccharic acid:- Saccharic acid, also called glucaric acid, is a chemical compound with the formula $C_6H_{10}O_8$. It is derived by oxidizing a sugar such as glucose with nitric acid.



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